

# Project #5: Autonomous Vehicle Interactions at an Intersection

## Scope of Work

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## Introduction

**Team Members:** Logan Baardson, Ryan Kramer, Nay Law Htoo, Arturo Garcia

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**Client:** Nebraska Department of Transportation (NDOT)

Nebraska is getting ready for self-driving cars on its roads. The Nebraska Department of Transportation (NDOT) wants to see real proof of how these cars perform. Even though people say self-driving cars are safer than humans, we need to look at the hard data to see how they handle tricky city streets with unpredictable drivers, walkers, and bikers.

## Project Goals

The goal of this project is to be able to evaluate the claim that AVs (Autonomous Vehicles) are safer and more predictable than others, such as pedestrians, bicyclists, etc. This team will analyze 200 seconds of trajectory data that was provided from the TGSIM (Third Generation Simulation) dataset, more specifically from the Foggy Bottom segment, to be able to study behavior and any interactions happening around intersections.

## Technical Approach and Task

To achieve the goals, the following key tasks have been identified below to complete.

### Task 1: Create a Gantt Chart

A Gantt chart will be created, which includes all the project tasks for which team members are responsible for each task, and a timeline for each task to be completed. This chart will allow the team to keep on schedule and complete all tasks by the project deadline.

### Task 2: Engineering Timesheet

The team will keep a detailed log of the hours worked on the project. This timesheet will show exactly what each person did and how much time they spent on specific tasks. It's a way to prove that the work was distributed fairly and that every part of the project received the attention it needed.

### Task 3: Write The Code

This is the phase where we build the actual tool. We will write a computer program using a Jupyter Notebook and the given data. We will use the steps below to write the code to find an answer to the client's questions.

#### *Step 1: Clean Dataset*

The data we're getting is probably not organized yet. We need to use Python to organize it, get rid of any weird errors, and make sure we're looking at the right cars and pedestrians.

#### *Step 2: Using tools*

We aren't building everything from scratch. We will install three specific toolkits (packages) to help us handle the calculations and graphs.

#### *Step 3: Set functions*

We're going to write a specific function(s) that automatically calculates things like how fast a car slams on its brakes, or if it stays at a steady speed, graphs, and biases/ethics analysis.

#### *Step 4: Create charts*

We're going to make at least two big graphs if not, more than 2 at least. One will probably show a "map" of the intersection with dots moving around, and the other will compare things like the average speed of a human driver versus a self-driving car.

## Task 4: Annotated Code Document

We also need to explain how the code works. We will create a document that breaks down the code line-by-line. This annotated guide will explain what each section of the code is doing so that someone else can easily understand and use it later.

## Task 5: Final Report

We will put all our findings and results into one main document. This report will explain the entire project from start to finish, including what we discovered about self-driving cars. It will use simple tables and charts to show the Nebraska Department of Transportation exactly what the data means. The report will also talk about the technical summary, which will include detailed methodology, function definitions, and statistical significance of AV behavior, and the nontechnical summary, which will include High-level insights for NDOT stakeholders, focusing on whether AVs met the "predictability" claim. The report will also include Bias/Ethics Analysis. A specific write-up that talks about the "fairness" of our code. For example, if our code ignores pedestrians, we must explain why that's a problem or a bias in our study. Finally, the report will also discuss the prompt the client asked for, answering them using the data we found.

## Task 6: GitHub

All our work, the code, the data, and the instructions will be uploaded to GitHub. We will include a "README" file, which acts like a front cover or summary for the project. This makes it easy for the client to find, download, and review everything we produce in one digital folder.

## Deliverables

The project will produce several key deliverables for the client and will be delivered on time.

| <b>Deliverable</b>                                                                                            | <b>Due to Client By</b> |
|---------------------------------------------------------------------------------------------------------------|-------------------------|
| Cleaned and raw CSV datasets                                                                                  | 05/08/2026              |
| Jupyter Notebook (.ipynb) with code                                                                           | 05/08/2026              |
| Gantt Chart                                                                                                   | 05/08/2026              |
| Annotated Code Document                                                                                       | 05/08/2026              |
| Written technical report summarizing methods, results, bias/ethics analysis, interpretations, and discussions | 05/08/2026              |
| Engineering Timesheet                                                                                         | 05/08/2026              |
| Scope of Work                                                                                                 | 05/08/2026              |

|                                         |            |
|-----------------------------------------|------------|
| GitHub Repository (link with all files) | 05/08/2026 |
|-----------------------------------------|------------|

## References

- [Transformed\\_TGSIM\\_Foggy\\_Bottom\\_200sec.csv](#)
- [TGSIM\\_Data\\_Dictionary.xlsx](#)
- [Polygons.ipynb](#)